

Designing a Music Tour: Planning and Creating a Profitable Concert Tour through Data Analysis

Summary

In the current market and entertainment industry, securing a successful concert is crucial for the future of a musical performer. Ticket sales, impact brought on the local economy, fan attendance, and many other factors can all be considered to plan a successful tour for the musical performer. To plan a successful concert for the picked musician Justin Bieber, we brainstormed many factors and issues that may appear when it comes to managing a tour.

By researching for data about the “**Purpose Tour**”, we planned the locations for the concerts based on these data and used them for future planning.

We also defined variables like **ticket price**, **fan attendance**, and **local economy**, and calculated the percentage each variable takes up for the whole. On top of that, we created a mathematical model based on the idea of finding the relationship between the ticket price and the **local GDP**, the relationship between the estimated ticket price and the number of tickets sold. In addition to that, we found the best way to maximize revenues and fan attendance through finding an equation and showing how the graph of the equation reflects how to achieve maximum profit. Then, we searched graphs and data relating to the local economy of hosting cities. Finally, we concluded that **high attendance rates** combined with **high ticket prices** were the greatest strengths of the concert. We also found that the most suitable ticket price is **\$90**, and there should be **130 concerts** in the tour. Our model suggested that the musician should consider the **GDP per capita** of hosting cities in future tours.

We separated the cities into the **3 tiers**: tier 3, tier 2 cities, and tier 1. In 130 concerts, there are **13** concerts held in a tier 1 city, **26** concerts in a tier 2 city, and **91** concerts in a tier 1 city, which shows that there will be **108** host cities.

The total money earned in a tour without subtracting the cost is **\$502,320,000**.

We summarized our ideas and wrote a letter to the musician. We then thought about how our plans could influence more musicians and why the model could not be used for every performer.

Keywords: Ticket Price, Fan Attendance, Local GDP, Local GDP per capita



Contents

1. Introduction	3
1.1 Problem background	3
1.2 Restatement of problem	3
2. Assumptions	4
3. Notations	4
4. The Concert Simulation Models	4
4.1 The relationship between selling price and GDP	5
4.2 Concert stadium capacity	7
4.3 Concert location planning with consideration of GDP	8
4.4 The route planning	16
5. Plan for Future	17
6. Strengths & Weaknesses	18
7. Letter to Justin Bieber	19
References	21



1 Introduction

1.1 Problem background

Music is an important factor in entertainment and communication between different cultures. It leads to the conveying of different ideas between diverse groups, merging people of varied backgrounds and heritage into a wide group with similar interests and hobbies [1-3]. The popular artists of the time have millions of fans, and people go to their tours and concerts, spending great expense on transportation, tickets, and merchandise, thus increasing the revenue of the artists.

As influential musicians go on tours, even if their main purpose isn't the profits, they still make significant amounts of money, even with the cost, taxes, and donations subtracted [4]. The musicians plan their tours clearly, as the route, quantity, and location of concerts all take-up time and energy, and a good plan can keep the musician's physical and status fine and the fans satisfied, thus having a positive impact on the artist's reputation; however, a bad concert plan and cancelled concerts can lead to the negative reflection of the artist him/herself. As most successful concerts have approving critique and high profits, it could be defined as what a successful concert is compared to mediocre or poor-quality ones.

Justin Bieber is a distinguished musician that has more than 100 million followers. His concerts are always with full attendance, and the revenue pulled in from one concert is always at least 1 million dollars. **Analyzing the past records** of his concerts, a model could be made to establish a superior plan for a concert concerning the highest profits and giving him a better reputation.

1.2 Restatement of problem

In this problem, we are asked to help a musician plan a future concert tour by creating a model that uses the past revenues from the shows in each concert and the local economy in the host cities to determine which country and city is the most suitable for opening more concerts and increase the profit to a maximum degree [5]. To achieve these goals, we need to:

-**analyze the local economy** of each host cities (especially the cities that hosted several shows in a single tour).



- finds the fan numbers** in each city and their consumption capacity.
- create a model that lists the economy of host cities, the fans' consumption capacity, and their relationship with the profit made.
- finds the most important factor that determines the revenue of the tour.
- determine the **best price** on a global scale.
- plan the future tour**, including cities.

2 Assumptions

To simplify our modeling, we make the following assumptions:

Assumption 1 We assume that the **cost for each concert of the tour is the same**. Due to the large amounts of profit generated by each concert, we can assume that the cost for each concert is one million dollars. This makes sense for the amount of money needed to rent the stadium, pay for travelling around different places, and so on. Since this number doesn't vary, we would use the same value.

Assumption 2 We can assume that the **capacity of the stadiums is 20,000 thousand people, 30,000 people, and 50,000 people**. We can divide the cities of the tour into **three tiers** according to their local GDP.

3 Notations

- A fan attendance
- P Ticket price
- G GDP contribution
- D_i stadium capacity of the i -th concert
- E local economy
- S_i profit of the i -th concert
- T total profit of a tour
- N number of concerts
- C_i the cost of the i -th concert

4 The Concert Simulation Models

The number of reported results varies from day to day, and analyzing the changes in this data can, to some extent, reflect the relationship between different



variables and the revenues of the concert tours. By studying historical data and trends, it may be possible to make informed predictions about future reported results.

To open a tour, many factors, such as fan attendance, the GDP of a concert location, or ticket prices, need to be considered [6, 7]. To determine the best locations for opening Bieber's concerts, a model can be built to decide on the preferable range of the ticket prices of each concert, and the GDP of the cities where said concerts are held.

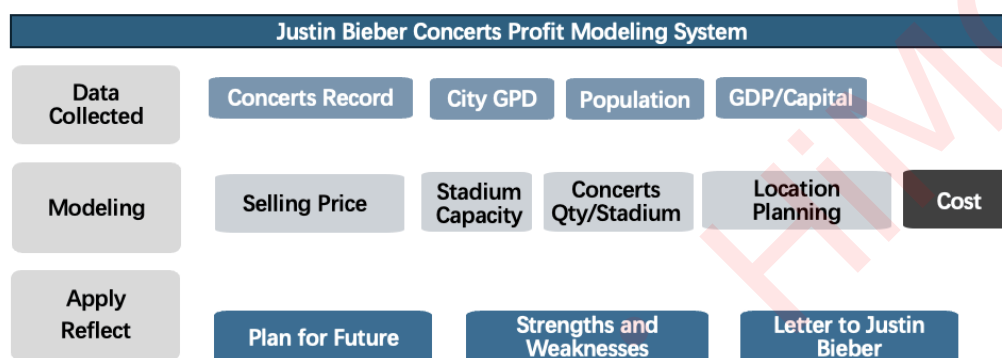


Figure 1. Schematic diagram of Justin Bieber concerts profit modeling system.

We analyzed different data about Justin Bieber's past tours [8]. Since the Purpose World Tour is the tour that Justin Bieber made the most money and our goal is to maximize the profit of our tour, most of our references to past data will be about the **Purpose World Tour**.

4.1 The relationship between selling price and GDP

The relationship between the price of each ticket and the GDP of the specific location is directly proportional. On the graph right here, the **X axis is the price per ticket, and the Y axis is the GDP per capita of the host cities**. The host cities are the **64 cities** that participated in the Purpose Tour.

By finding out the sum of money earned and the amount of people attending the concert, we were able to find out the average price for each single concert. Then, we **researched the GDP per capita for each city** [9].

With all of the data ready, we were able to make a graph to demonstrate the **relationship between the ticket price and the GDP per capita of the hosting city**.



The graph shows that the ticket price increases as the GDP per capita increases. To the point that the price and GDP per capita reaches a maximum, it starts to decrease gradually.

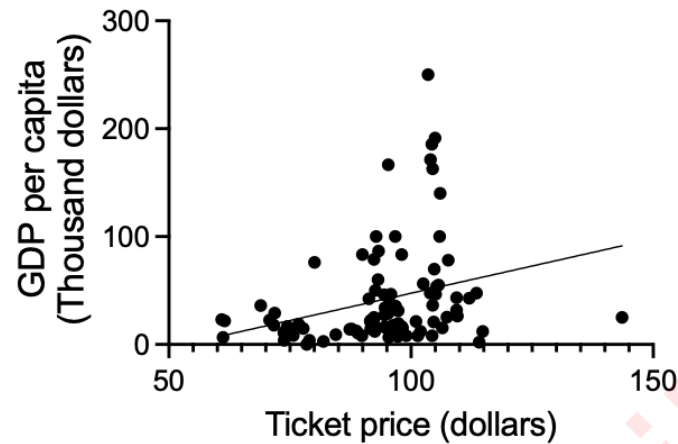


Figure 2. Relationship between the ticket price and the GDP per capita of the hosting city

We examined the data from the "Purpose Tour" and found out that most of the concerts had a fan attendance rate of 100%, meaning that all the fans came to the tour. As for some other areas, the fan attendance rate was less than or equal to 90%. Most of these cities were in developing countries in Latin America and developed cities in relatively small countries in Northern Europe. There are two possible explanations:

- i. **Developing countries** have relatively low GDP per capita, thus some citizens are unable to pay for the relatively high price of the ticket. This causes the fan attendance rate of these concerts to decrease. For example, the fan attendance for Lima, Peru, is 85.49%.
- ii. Justin Bieber's fame is **mostly based on North American** countries and certain countries in Europe. The number of fans in Northern Europe is less compared to the number of fans in other areas. Thus, the number of tickets sold may be less compared to other areas.

This indicates that the **ticket price is somehow correlated to the GDP per capita** of the region, meaning that the local economy still plays a rule in the setting of the price.

Therefore, we developed a model to find a new correlation.



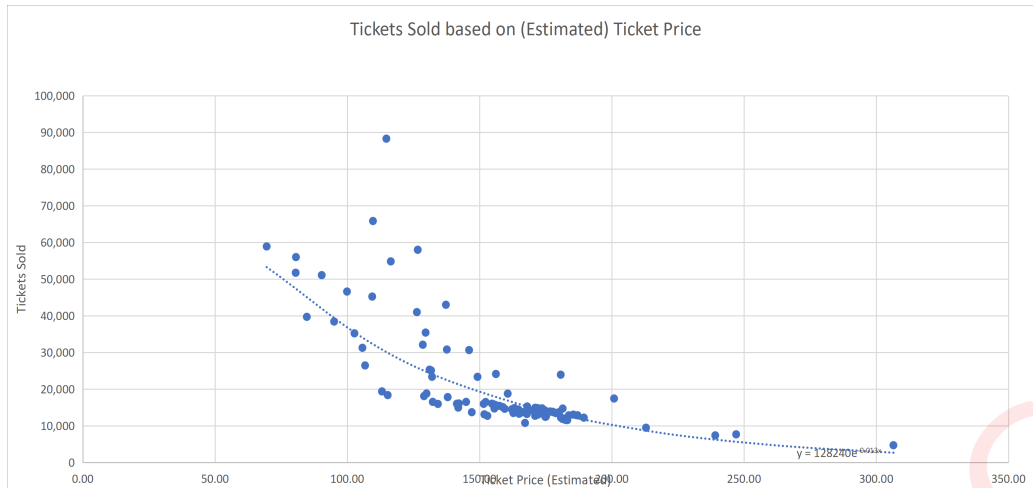


Figure 3. The relationship between the tickets sold and the estimated ticket price.

From the graph, it is seen that there is a direct relationship between the ticket price and the fan attendance rate. After using Excel, the curve fitting formula for this graph is $y = 128240e^{(-0.013x)}$, where x is the ticket price estimation, and y is the number of tickets sold.

The main revenue of the concert is the product of the number of tickets sold and the price of the ticket. Using the equation above, the total revenue can be seen as $y = 128240e^{(-0.013x)}x$

Using this equation, the graph below can see:

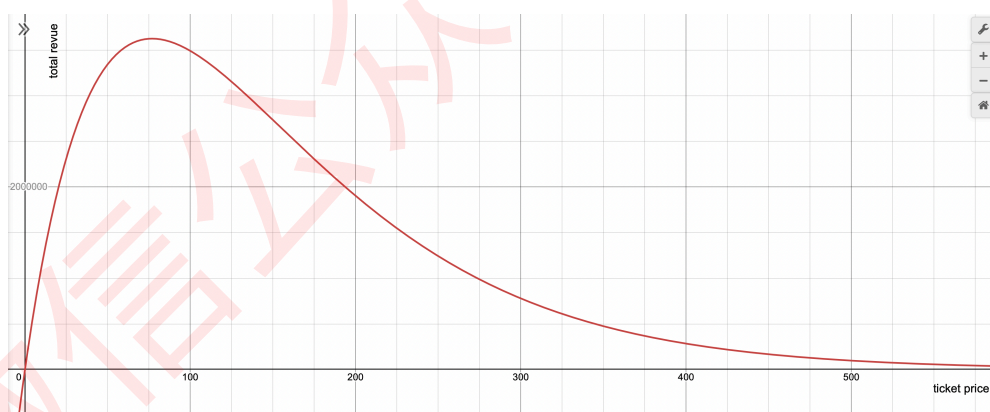


Figure 4. Forecast graph for revenue maximization.

Looking at the graph, the price of a concert ticket to maximize the revenue is \$92. Thus, the price for his concert tickets should be \$92.

4.2 Concert stadium capacity



Since we assume that the stadium capacity is 20,000 people, 30,000 people, and 50,000-80,000 people. We now need to know how many cities have a 50,000-person stadium.

In general, we **excluded the cities with low GDP** or those that **aren't spacious enough**. There won't be many concert performances in a minimum-standard stadium. 40% of the stadiums that we will be using are of maximum standard, while the other 40% are of medium stadiums. The remaining 20% are of the minimum stadium. Therefore, the average capacity of every concert is: $40\% * 65,000 + 40\% * 30,000 + 20\% * 20,000 = 42,000$

Table 1. The stadium capacity

Stadium	Capacity/Concert
Maximum	50,000-80,000
Medium	30,000
Minimum	20,000

4.3 Tours and concert location planning with consideration of GDP

Considering the fact that Bieber's physical condition may deteriorate due to his increase in age, a **12-month tour** may be more suitable and beneficial for his health in contrast with his 15-month-long tour in the past.

Based on Justin Bieber's tour history, his Purpose World Tour has 162 concerts over an expansion of 15 months. **In a ratio of 12 months, the tour should have 130 concerts.**

We analyzed the data based on the number of Justin Bieber fans [8, 10] and found 18% of Justin Bieber's fans are in the US, 8% in Brazil, 4.25% in the UK, and 4% in Germany. Distributing the concert numbers to the countries according to fan distribution. The number of concerts in each country is listed in Table 2.

Table 2. The number of concerts in all countries investigated.

Country	Concerts	Country	Concerts
United States	23	Indonesia	3
Brazil	10	Australia	3



United Kingdom	6	Vietnam	2
Germany	5	Argentina	2
Mexico	5	Colombia	2
Canada	5	Peru	2
France	4	Venezuela	1
Turkey	4	Netherlands	1
Poland	3	Philippines	1
Spain	3	Malaysia	1
Italy	3	Indonesia	3

The countries with the highest fans of Justin Bieber are listed, leaving the rest 30% of the concerts to the other cities.

Based on the purpose of the tour, **countries with low populations or low economies should be eliminated**, thus producing the chart as shown in Table 3.

Table 3. The number of concerts in countries selected.

Country	Concerts	Country	Concerts
United States	26	Italy	2
Brazil	10	Vietnam	2
United Kingdom	6	Mexico	2
Germany	3	Argentina	1
Australia	3	Colombia	2
Indonesia	2	Peru	2
Canada	2	Venezuela	1
France	2	Netherlands	1
Turkey	2	Philippines	1
Poland	2	Malaysia	1
Spain	2	Remaining	55

The **cities can also be categorized into 3 tiers** with tier one cities having 3 concerts, tier two cities having 2 concerts, and tier three cities having 1 concert. The ratio of the three types of cities can be determined as **1:2:7**. As tier one cities require high GDP and population to hold three concerts, and the number of concerts are limited, it isn't practical to let many cities host three concerts.

During the Purpose World Tour, cities like London, Paris, and Birmingham held more amounts of concerts while most of the other cities only had one concert. **The cities that can support the three-concert-plan should have enough people who can afford to attend them.** Thus, the cities should be



ranked by the GDP and population and split into the three tiers to determine the number of concerts.

There is another way to reach this conclusion: we calculate the **relationship between the numbers of tickets sold and the GDP per capita for hosting cities**:

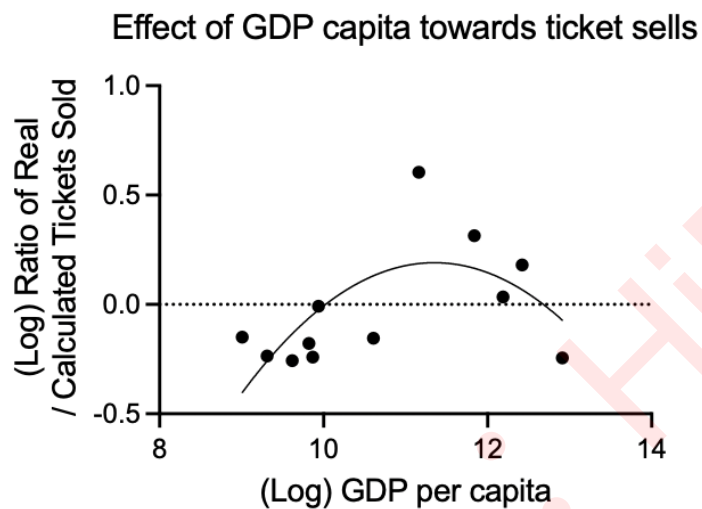


Figure 5. The effect of GDP capita on ticket sales.

As shown in Figure 5, the number of tickets sold increases as the GDP per capita increases until the GDP per capita reaches its maximum point of about $e^{11.5}$ dollars. This may be because, in cities that have high GDP, people might have higher expectations about happiness. Therefore, they are less likely to purchase concert tickets when compared to cities with a lower GDP. This correlation could be roughly measured by a quadratic function $f(x) = -0.107x^2 + 2.443x - 13.677$. X stands for the natural logarithm of the numeral representation of GDP in dollars, and the function returns the logarithm of the real data of tickets sold divided by the estimated number of tickets sold, computed using the previous Ticket Sells - Ticket price model.

In addition to the influence of GDP per capita, total city GDP also shares a positive correlation with ticket sales, as shown in this graph.



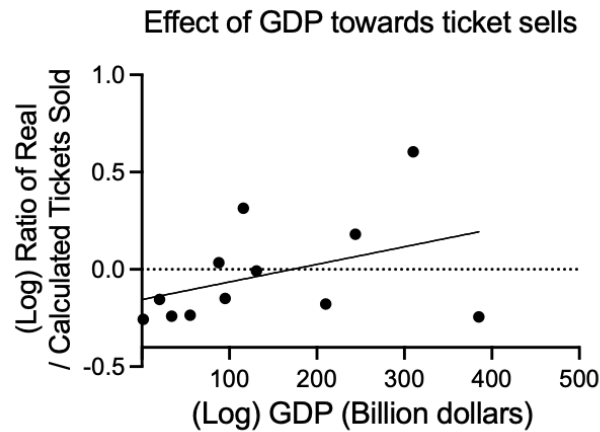


Figure 6. The effect of GDP on ticket sales.

This graph reflects that in general, higher city GDP brings more tickets sells. This may be caused by the convenience and increase of life quality brought by an increase of GDP. The relationship is estimated through a linear function $g(x)=0.0009x-0.155$ where x stands for that city's GDP in billion dollars, and the function returns the logarithm of real data of tickets sold divided by the estimated number of tickets sold, computed using the previous Ticket Sells - Ticket price model.

With these two pieces of data, we can see that ticket sells are influenced economically by two factors: GDP per capita and total GDP of that city.

Based on our analysis, the overall influence of economics is measured through the expression $T=0.7GDP_{capita}+0.3GDP_{total}$. GDP per capita is given a weight of 0.7, as ticket price depends more on individual choices.

This gives a mathematical way to measure the marketing effectiveness of a city based on its GDP and GDP per capita [9], with both sides presented in a logarithmic way:

$$\ln(M)=0.7f(GDP_{capita})+0.3g(GDP_{total})$$

Also, by using this equation, we can estimate the maximum profit of a single concert (using the optimal ticket price of 90 dollars):

$$Raw\ Income \approx (5500000\ dollars) \times e^{0.7f(GDP_{capita})+0.3g(GDP_{capita})}$$



The left side of the product represents the maximum raw income in an average city, as shown in the previous model. The right side represents the influence of GDP on the city where the concert is hosted.

After ranking the cities using this formula, we get the optimal list of concerts and their locations in the USA, Brazil, and the UK (Table 4-6).

Table 4. The concerts and their locations in the USA.

City	Population	GDP (million USD)	GDP/capita (USD)	No. of concerts
San Jose	1,952,185	410,418	210,235	3
San Francisco	4,623,264	668,677	144,633	3
Seattle	4,011,553	479,966	119,646	2
Boston	4,899,932	531,671	108,506	2
New York	19,768,458	1,992,779	100,806	2
Los Angeles	12,997,353	1,124,682	86,532	2
Denver	2,972,566	253,399	85,246	2
Austin	2,352,426	193,773	82,372	1
Minneapolis	3,690,512	296,969	80,468	1
Chicago	9,509,934	764,583	80,398	1
Baltimore	2,838,327	222,967	78,556	1
Atlanta	6,144,050	473,823	77,119	1
Dallas	7,759,615	598,333	77,109	1
Charlotte	2,701,046	207,866	76,958	1
Philadelphia	6,228,601	\$477,580	76,675	1
Houston	7,206,841	\$537,066	74,522	1
Miami	6,091,747	\$417,147	68,477	1

Table 5. The ten concerts in Brazil

City	Population (million)	GDP (billion USD)	No. of Concerts
São Paulo	12	462.23	3
Rio de Janeiro	6.21	161.33	2
Minas Gerais	20.54	145.69	2
Rio Grande do Sul	10.88	98.77	1
Paraná	11.44	93.33	1
Santa Catarina	7.61	72.76	1



Table 6. The six concerts in the UK

City	GDP (billion)	Population (million)	No. of Concerts
London	708.34554	8.87	2
Manchester	125.63964	0.56	1
Birmingham	107.34192	0.20	1
Leeds	93.7503	0.79	1
Liverpool	57.22038	0.50	1

There should be 3 concerts in Germany and Australia; 2 concerts in Canada, Mexico, Vietnam, Argentina, Columbia, Peru, Poland, Spain, Indonesia, and Italy; and 1 concert in Pakistan, Venezuela, the Netherlands, the Philippines, and Malaysia (Table 7).

Table 7. The concerts in other countries

Country	City	GDP (billion USD)	Population	No. of concerts
Germany	Berlin	271.818	4,750,000	1
Germany	Hamburg	152	4,750,000	1
Germany	Frankfurt	197.259	9,499,000	1
Australia	Sidney	398	5,400,000	1
Australia	Melbourne	318.677	5,050,000	1
Australia	Brisbane	177	2,388,038	1
Indonesia	Bali	29.901	2,621,877	1
Indonesia	Bandung	38.29	7,050,000	1
Canada	Toronto	306	2,790,000	1
Canada	Vancouver	135.6	662,250	1
France	Paris	934.168	11,400,000	1
France	Nantes	65.457	859,972	1
Turkey	Istanbul	275.365	16,500,000	1
Turkey	Ankara	74.83	5,150,000	1
Poland	Warsaw	135.343	2,375,000	1
Poland	Katowice	52.713	2,325,000	1
Spain	Alicante	38.851	1,915,282	1
Spain	Barcelona	182.524	4,775,000	1



Italy	Milan	233.596	6,200,000	1
Italy	Alicante	38.851	1,009,088	1
Vietnam	Hanoi	42.04	8,053,663	1
Vietnam	Ho Chi Minh City	96.399	10,900,000	1
Mexico	Monterrey	140	5,324,281	1
Mexico	Mexico City	340.8	24,700,000	1
Argentina	Buenos Aires	235.6	16,700,000	1
Colombia	Bogotá	121.8	10,200,000	1
Colombia	Medellín	35	4,225,000	1
Peru	Lima	114.4	11,600,000	1
Peru	Cusco	4.2	505,000	1
Venezuela	Caracas	70	1,942,652	1
Netherlands	Amsterdam	237.841	2,500,000	1
Philippines	Manila	162.3	26,400,000	1
Malaysia	Kuala Lumpur	59.831	8,900,000	1

Table 8. The remaining 54 concerts

City	Country/region	GDP (billion USD)	Population	No. of concert
Greater Tokyo Area	Japan	2080.76	40,400,000	4
Seoul Capital Area	South Korea	878.54	24,800,000	3
Kyoto–Osaka–Kobe	Japan	699.47	17,700,000	3
Shanghai	China	663.87	24,890,106	4
Beijing	China	618.65	20,500,000	3
Singapore	Singapore	530.71	5,938,000	2
Moscow metropolitan area	Russia	504.81	17,400,000	2
Shenzhen	China	455.69	14,678,000	2
Guangzhou	China	429.00	18,676,000	2
Nagoya	Japan	379.30	10,500,000	2
Tel Aviv metropolitan area	Israel	310.00	2,775,000	2
Abu Dhabi metropolitan area	United Arab Emirates	277.93	1,740,000	2
Greater Dublin Area	Ireland	272.45	1,420,000	2



Busan-Gyeongnam Area	South Korea	252.15	5,115,000	2
Bangkok Metropolitan Region	Thailand	252.13	19,900,000	2
Riyadh	Saudi Arabia	243.79	8,800,000	1
Amsterdam metropolitan area	Netherlands	237.84	2,500,000	1
Brussels metropolitan area	Belgium	237.08	2,725,000	1
Milan metropolitan area	Italy	233.60	6,200,000	1
Stockholm metropolitan area	Sweden	203.00	2,250,000	1
Dubai-Sharjah-Ajman metropolitan area	United Arab Emirates	202.80	6,050,000	1
Greater Cairo	Egypt	190.20	21,500,000	1
Fukuoka–Kitakyushu	Japan	182.95	4,750,000	1
Copenhagen metropolitan area	Denmark	179.85	1,690,000	1
Rome metropolitan area	Italy	175.34	3,575,000	1
Vienna	Austria	165.97	2,250,000	1
Zürich Metropolitan Area	Switzerland	158.91	1,430,000	1
Dhaka	Bangladesh	145.00	20,900,000	1
Monterrey	Mexico	140.00	5,324,281	1
Auckland	New Zealand	139.50	1,531,400	1
Jeddah	Saudi Arabia	138.97	5,300,000	1
Warsaw metropolitan area	Poland	135.34	2,375,000	1
Johannesburg	South Africa	134.68	8,400,042	1



4.4 The Route Planning

One thing illogical about the Purpose World Tour is that the concert location changes need Bieber to take long-term flights often, and some cities such as London have concerts for multiple times, yet they aren't in a row, but Bieber had flown from and landed many times. During this tour, the route of the cities should be strictly zero-wayed, and a city shouldn't be toured twice. This route lets the minimum amount of time to be wasted on transportation and jet lag.

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5 Plan for Future

In conclusion, we are able to propose the following plan for Justin Bieber's next concert tour.

In the 130 concerts in Justin Bieber's next tour, the concerts should be ranked like this:

In USA:

-3 concerts in San Jose and San Francisco

-2 concerts in Seattle, Boston, New York, Los Angeles, and Denver

-1 concert in Austin, Minneapolis, Chicago, Baltimore, Atlanta, Dallas, Charlotte, Philadelphia, Houston, and Miami.

In the other parts of North America:

-5 concerts in Mexico and Canada

In Brazil:

-2 concerts in São Paulo, Rio de Janeiro and Minas Gerais

-1 concert in Rio Grande do Sul, Paraná, and Santa Catarina

In the other parts of South America:

-2 concerts in Argentina, Columbia, and Peru

-1 concert in Venezuela

In Europe:

-6 concerts in the UK

-5 concerts in Germany

-4 concerts in France and Turkey

-3 concerts in Poland and Spain

-1 concert in the Netherlands

In Asia:

-3 concerts in Indonesia

-2 concerts in Vietnam

-1 concert in Pakistan, , the Philippines, and Malaysia

In Oceania:

-3 concerts in Australia

In Africa:

-2 concerts in South Africa

The remaining 40 concerts can be in any country that is profitable.



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6 Strengths and Weaknesses

Strengths

The beneficial part of our model is that, after researching different data, our model is **the closest** to an ideal solution. Our model can be used to help many musicians plan concert tours that may be able to maximize profit for the musician. Generating more profit is a crucial sector of We can access that because we searched information on the GDP per capita and the created graphs that clearly shows the relationship between our different factors.

Weaknesses

However, our model consists of problems such as:

- other musicians **may not have the fame** of Justin Bieber.
- the model is **idealistic; some factors, such as routes and transportation**, weren't included in our model.
- the **cost** is based on an assumption. In a **real-life** situation, the cost of a single concert may vary according to location.

Will our model be completely useful for every musician?

Not quite-- our model can be used for most artists when planning their concerts. Since our targeted artist is Justin Bieber, a famous Canadian singer, our model is **more suitable for performers who are famous**. After many considerations, we can conclude that only reputable artists like Justin Bieber are able to have such high fan attendance rates to the point that most of them are 100%.



7 A Letter to Justin Bieber

Dear Justin,

We hope this letter finds you well. As your tour managers at MidMCM Music Agency, we are excited to present a comprehensive tour model that we have developed to optimize your upcoming concert tour. Our team has analyzed your past tours and identified key factors that contribute to a successful and profitable tour. We believe that our model will not only meet but exceed your goals for fan engagement and financial success.

Our Approach:

- Familiarization with your **touring history**.
- We began by examining your previous tours, focusing on the reasons behind your tours, duration, number of shows, and locations. This analysis provided us with a solid foundation to understand your touring patterns and fan base.
- Identifying key factors
- We considered various factors that impact a single show and the overall tour, including **logistics, venue capacity, ticket pricing, and promotional strategies**.

Model development

- Our model incorporates a dynamic approach to planning, taking into account the number of concerts, selection of venues, and ticket pricing strategies. We have also integrated a module to predict fan attendance based on historical data and market trends.

Maximizing Goals

- We have designed the model to help achieve various objectives, such as **maximizing fan attendance and optimizing profits**. The model suggests strategies tailored to your specific goals and the unique dynamics of each tour.

Our Recommendations:

- City Selection: Based on our analysis, we recommend **targeting cities with a high concentration of your fan base** and a track record of successful past shows. We separated the cities into the **3 tiers**: tier 3 cities will hold 3 concerts, tier 2 cities will hold 2 concerts, tier 1 cities will hold one concert. After using our model, we concluded that in 130 concerts, there are **13** concerts held in a tier 1 city, **26** concerts in a tier 2 city, and **91** concerts in a tier 1 city, which shows that there will be **108** host cities.
- Ticket Pricing: To maximize both attendance and revenue, we propose that the best value for **ticket price is \$90**.

Why This Tour Will Be a Success:



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Our model predicts that by following these recommendations, the profit of your tour before subtracting the cost (such as your transportation, venue price, and staff) of the concert is **\$502,320,000**. The tour will not only be **financially successful** but also create **lasting memories** for your fans. We believe this approach will enhance your brand, expand your reach, and solidify your position as a top-tier performer in the music industry.

Looking Forward:

We are eager to discuss this model in more detail and tailor it to your specific needs and vision for the tour. We believe that with this strategic approach, your upcoming tour will be your most successful yet. Please feel free to reach out to us with any questions or to schedule a meeting to further discuss the details. Thank you for entrusting MidMCM Music Agency with this exciting opportunity.

Sincerely,

Our team,

#15637



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